Unit 2 — Exponents & Scientific Notation

★ Answer Key ★

Do not distribute to students until after the lesson.

Guided Notes — Fill-Ins

§1 Product & Quotient. Product: $x^3 \times x^4 = x^7$; $(2x^3)(3x^2) = 6x^5$; “Same base? ADD.” Quotient: $x^6 \div x^2 = x^4$; $(12x^7) \div (3x^2) = 4x^5$; “Same base? SUBTRACT.”

§2 Power. Power: $(x^2)^3 = x^6$; $(2x^3)^2 = 4x^6$; “Power to power? MULTIPLY.” Power of a Product: $(2x)^3 = 8x^3$; $(3ab)^2 = 9a^2b^2$; “Give EACH factor the power.” Power of a Quotient: $(2/3)^2 = \frac{4}{9}$; $(a/b)^3 = \frac{a^3}{b^3}$; “Power to top AND bottom.” Parentheses: $-3^2 = -9$ vs $(-3)^2 = 9$.

§3 Zero & Negative. Zero: $5^0 = 1$; $(3x^5)^0 = 1$; “ANY base^0=1.” Negative: $2^{-3} = \frac{1}{8}$; $4x^{-3} = \frac{4}{x^3}$; “Flip to a fraction.” **Across the bar:** $\frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$. **Reciprocal:** $(\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$.

2.2 Scientific Notation. Large → +exp; Small → −exp. Multiply: MULTIPLY coefficients, ADD exponents, adjust to 1–10. Divide: DIVIDE coefficients, SUBTRACT exponents. Examples: $(3\times10^4)(2\times10^5) = 6\times10^9$; $(8\times10^6) \div (4\times10^2) = 2\times10^4$; $(5\times10^3)(4\times10^2) = 20\times10^5 \to 2.0\times10^6$.

Practice 1 — Expand & Condense

1) $x \cdot x \cdot x \cdot x \cdot x \cdot x$ 2) $4 \cdot a \cdot a \cdot a$ 3) $2 \cdot y \cdot y \cdot y \cdot y \cdot y$ 4) x^5 5) $3b^4$ 6) m^7 7) $(2x)(2x)(2x) = 8x^3$ 8) $(5a)^2 = 25a^2$ 9) $(xy)^4 = x^4y^4$ 10) $\frac{x^3}{y^2}$ 11) $\frac{2a^2}{b^3}$ 12) $\frac{m^2}{n^4}$

Practice 2 — Combining Like Terms

1) $8x$ 2) $5y$ 3) $8a^2$ 4) $4mn$ 5) $9x^2y$ 6) $7p$ 7) $11c^3$ 8) $5x + 3y$ (cannot combine) 9) $2x^2 + 4x$ (cannot combine) 10) $4a^2 + 3a$ 11) $2m^2 + 7m$ 12) $10a - 2b$

Practice 3 — Product & Quotient

1) x^{11} 2) $15x^8$ 3) $8a^5b^7$ 4) $-18x^5y^4$ 5) $30m^3n^6$ 6) x^5 7) $5a^5$ 8) $4x^3y^4$ 9) $4m^5n^2$ 10) $-6p^4q^3$ 11) $3x^5$ 12) $2a^4b^2$

Practice 4 — Power Rules

1) x^{20} 2) $16x^{12}$ 3) $-27a^6$ 4) x^6y^{15} 5) $16m^6n^4$ 6) $32x^{15}y^{20}$ 7) $\frac{x^8}{y^{12}}$ 8) $\frac{8a^3}{b^6}$ 9) $\frac{9x^8}{4y^2}$ 10) x^{24} 11) $72x^8y^7$ 12) $16a^{12}b^8$

Practice 5 — Zero & Negative

1) 1 2) 1 3) $\frac{1}{64}$ 4) $\frac{2}{x^5}$ 5) $\frac{y^2}{x^3}$ 6) $\frac{b^3c}{a^2}$ 7) $\frac{4n^5}{m^6}$ 8) $\frac{4y^3}{x^4}$ 9) $\frac{y^2}{x^2}$ 10) $\frac{27}{8}$ 11) $\frac{b^3}{a^6}$ 12) $\frac{y^4}{9x^2}$

Practice B — Conversions

1) 4.5×10^7 2) 3.2×10^{-4} 3) 602,000,000 4) 0.000091 5) 7.8×10^5 6) 6.7×10^{-6} 7) 34,500 8) 0.0012 9) 5.28×10^9 10) 0.025 11) 9.04×10^{-7} 12) 7,150,000 13) 6.34×10^{10} 14) 0.0000088

Practice C — Operations

1) 6×10^9 2) 2×10^4 3) 3.5×10^{10} 4) 3×10^3 5) 3.0×10^{10} 6) 6×10^4 7) 4×10^{11} 8) 2.0×10^4

Practice D — Word Problems

1) $(9.3\times10^7) \div (1.86\times10^5) = 5\times10^2$ s (500 s) 2) 40× longer 3) 1×10^5 = $100,000 4) 5.2×10^{15} g 5) 3.7×10^8 m 6) 3×10^2 s (300 s) 7) $(2\times10^{12}) \div (4\times10^6) = 5\times10^5$ photos 8) $(3\times10^8)(6\times10^3) = 1.8\times10^{12}$ m

Cumulative Review

A · Exponent Rules: 1) x^{10} 2) $12x^7$ 3) $10a^4b^6$ 4) $4x^4$ 5) m^{15} 6) $16x^8y^{12}$ 7) $6a^3b^4$ 8) $\frac{x^{12}}{y^6}$

B · Conversions: 1) 7.2×10^6 2) 4.5×10^{-4} 3) 30,500,000 4) 0.000061 5) 9.1×10^4 6) 8×10^{-7} 7) 4,440,000 8) 0.0027

C · Operations: 1) 9×10^9 2) 4×10^6 3) 2×10^{10} 4) 6×10^6 5) 5.6×10^8 6) 4.8×10^2

D · Word Problems: 1) 4× wider 2) 5×10^3 h (5,000 h) 3) 1×10^6 widgets 4) $(6\times10^{-5}) \div (3\times10^{-7}) = 2\times10^2$ = 200× larger

E · Negative Exponents: 1) $\frac{1}{25}$ 2) $\frac{3}{x^4}$ 3) $\frac{y^3}{x^5}$ 4) $\frac{b^3}{a^3}$ 5) $\frac{5b^6}{a^5}$ 6) $\frac{y^2}{4x^4}$

F · Expand, Condense & Like Terms: 1) x^6 2) $(3a)^2 = 9a^2$ 3) $\frac{m^3}{n^2}$ 4) $11x$ 5) $5a^2$ 6) $5mn + 2m$